MILITARY SPECIFICATION

MIL-C-10879D(SigC)
22 July 1958
Superseding
MIL-C-10879C
5 April 1955

RADIOSONDE BASELINE CHECK SET AN/GMM-1( )
(Components of)

1. SCOPE

1.1 This specification covers the following components of Radiosonde Baseline Check Set AN/GMM-1( ): (See 6.3).

Temperature - Humidity Chamber
Control Unit
Carrying Case with Cover
Spool
Power Cable
Remote Control Cable

The set consists of a transportable temperature - humidity chamber with accessories for preflight testing of balloon - borne radiosondes, AN/AMT-2, and AN/AMT-4.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

LLL-F-291	Fiberboard; Corrugated, Single-Face (Flexible)
PPP-B-621	Boxes; Wood, Nailed and Lock Corner
PPP-B-636	Boxes; Fiber
PPP-T-76	Tape; Pressure-sensitive Adhesive; Paper, Water Resistant
UU-T-111	Tape; Paper, Gummed (Sealing and Securing)
QQ-S-571	Solder; Soft (Tin, Tin-Lead and Lead-Silver)

MILITARY

MIL-P-116	Preservation; Methods of
MIL-B-117	Bags; Interior Packaging
MIL-T-152	Treatment; Moisture-and-Fungus-Resistance of Communications, Electronic, and Associated Electrical Equipment

FSC 6660

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SPECIFICATIONS (CONTD)

MILITARY

MIL-V-173	Varnish; Moisture-and-Fungus-Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes; of Electronic Items

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycles for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

DRAWINGS

SIGNAL CORPS

SC-D-17269	Caution Plates
SC-DL-21271	Calibration Chamber ML-387( )/GM and Radiosonde Baseline Check Set AN/GMM-1( )
SC-B-61578	Grounding to Chassis
SC-A-82759	Name and Caution Plates
SC-C-33073	Form to be Used for Pilot Pack

(Copies of specificatinns, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification No. 4

(Application for copies should be addressed to: The Association of American Railroads, 202 Chicago Union Station, Chicago, 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification, Numbers A-3 and 14 for Truck Shipment.

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(Application for copies should be addressed to: The American Trucking Association, 1424 16th Street, N.W., Washington 6, D.C.)

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules 21, for Rail Shipment

(Application for copies should be addressed to: The Consolidated Freight Classification Committee, 1 Park Avenue and 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Description. - Radiosonde Baseline Check Set AN/GMM-1( ) consists of a plywood chamber provided with a closed working space for preflight testing of radiosondes. The temperature-humidity chamber provides an enclosed space in which stable conditions of temperature and humidity are created. The equipment has means for mounting on the floor, table or wall. A blower and blower motor provide adequate air circulation when making temperature and humidity measurements. A strip heater and tray of water are mounted within the chamber to expedite stabilization of temperature and humidity conditions. Various electrical connections terminating in a control unit enable manual or automatic "switching in" of the desired radiosonde circuits (i.e. reference, temperature and relative humidity.) A carrying case of reinforced plywood is provided for transit and storage of the checkset and its components.

3.2 Construction. - The equipment shall be constructed in accordance with Drawing SC-DL-21271.

3.3 Preproduction samples. - The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract.

3.4 Service conditions. - The equipment shall meet the following service conditions:

3.4.1 Equipment nonoperating. - The equipment shall comply with the operational requirements of 3.4.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit;

a. Temperature. - Continuous exposure in the range of +160° F to -80° F; However, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours.

b. Relative humidity. - Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet specification requirements.

c. Elevation. - Elevation up to 25,000 feet above sea level.

d. Orientation. - Indefinite storage in any orientation.

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e. Vibration and bounce and shock. - As described in 3.5.

3.4.2 Equipment operating. - The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combination likely to be encountered during worldwide operations:

a. Operation - Continuous operation in any specified manner, for a period of 1 year, alternating 2 hours on and 2 hours off, with no more than normal maintenance and replacement of parts. Compliance will be determined by examination of the preproduction sample.

b. Elevation. - Elevation up to 10,000 feet above sea level.

c. Relative humidity. - Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

d. Temperature. - Ambient temperature in the range of +150° F to -40° F. (The 150° F temperature includes effect of sun load). Exposure at high temperature extreme not to exceed 72 hours.

3.5 Vibration, bounce, and shock. -

3.5.1 Vibration. - Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second.

3.5.2 Bounce and shock. - The equipment shall meet the requirements of Table I.

Table I - Bounce and shock

Requirements	Inspection Paragraph	Performance after inspection
Bounce; light equipment	4.9	Full specification performance No physical damage except surface abrasion.
Shock; drop	4.11	Equipment shall be operable. Minor mechanical damage is permissible.

3.5.3 Preconditioning. - The equipment shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.6 Tropicalization. -

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3.6.1 Treatment of assembled equipment. - Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

(2) It does not incorporate any material listed in 3.6.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.6.2 Treatment of materials. -

3.6.2.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.6.2.2 Toxicity. - Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.6.2.3 Flexibility. - Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.6.2.4 Statement of treatment. - The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL-T-152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.6.3 Materials not resistant to moisture and fungi. - The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated
(b) Cotton
(c) Cork
(d) Felt, hair or wool
(e) Jute
(f) Leather
(g) Linen
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

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3.7 Finish, protective. - The equipment shall be given protective finish in accordance with Specification MIL-T-14072.

3.8 Mounted hardware. - After hinges, catches, screws, nuts, and the like have been mounted, any break in the film of paint, either on the hardware or the mounting surface, shall be touched up to provide a continuous protective coating. A reasonably exact color match between the painted surfaces and the touch-up coating shall be provided.

3.9 Marking. -

3.9.1 General: Marking shall conform to Specification MIL-M-13231, except that the notation "US" shall be replaced with "Signal Corps U.S. Army" on the nameplate data for articles and equipment.

3.9.2 Visibility. - Wherever practicable, parts and assemblies shall be mounted so that their identification markings will be readily visible with minimum disassembly of the equipment.

3.9.3 Serial numbers. - The following articles shall be serial numbered:

Temperature - Humidity Chamber

Carrying Case with Cover

3.10 Technical manual caution notice. - A technical manual caution notice conforming to Drawing SC-D-17269 shall be provided on each component that has operating controls. The notice shall appear on the operating panel, or in a prominent place if there is no operating panel. The caution notice and nameplate data may be combined as shown on Drawing SC-A-82759.

3.11 Cleaning. - Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.12 Wiring and cabling. -

3.12.1 Slack. - Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.12.2 Protection. - Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges, under specified service conditions. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes or slots in metal (Partitions, shields, and the like), they shall be protected by suitable grommets or bushings.

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3.12.3 Support. - Wires and cables shall be properly supported and secured to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equipment has been subjected to specified service conditions or has been serviced or repaired in a normal manner.

3.12.4 Clearance. - Clearance between wires or cables, and parts such as relays, resistors, and motors, shall be sufficient to avoid deterioration of wires or cables because of the heat dissipated by such parts under specified service conditions for the equipment. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be at least 1/4 inch unless use of specified parts makes such clearance impracticable.

3.12.5 Splicing. - Wires in a continuous run between two terminals shall not be spliced during assembly of the equipment.

3.12.6 Connections. - Terminal lugs for attachment to wire leads shall be such as to enable the equipment to meet specified performance and service requirements. Terminal lugs that grip the wire insulation shall be used where the wires are subject to breakage at the connection. Solder type terminal lugs, and the bared ends of leads for attachment thereto, shall be tinned, silver-plated, or lead alloy coated. Solderless terminal lugs shall be installed with tools recommended by the manufacturer of the lugs. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173.

3.12.7 Grounding. - Ground connection to mechanical parts, except the chassis, or frame, shall not be made to complete electrical circuits but only to eliminate high-potential ac points.

3.12.7.1 Grounding to chassis. - Ground connection to the chassis shall conform to Drawing SC-B-61578.

3.13 Soldering. -

3.13.1 Solder. - Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571.

3.13.2 Flux and cleaning agents. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.13.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or under-heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment.

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Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.14 Technical literature, tools and running spare parts. - Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (See 6.2)

3.15 Interchangeability. - Corresponding components and replaceable subassemblies and parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts to be considered replaceable shall be as determined by the contracting officer prior to submission for acceptance.

3.16 Detail requirements

3.16.1 Performance. -

3.16.1.1 Remote control selector mechanism. - A one R.P.M. motor driving a cam shall be utilized in the control unit to "switch in" the various radiosonde circuit in the following sequence; reference, temperature, reference and humidity. The switch shall be such that each circuit shall be operative for a period of not less than 4-1/2 seconds or more than 15-1/2 seconds as the cam closes or opens the respective microswitches.

3.16.1.2 Ventilation. - The air within the chamber with the door closed, shall be at a speed not less than 3 miles per hour across the floor, and not less than 10 miles per hour across the ceiling of the chamber. There shall be a minimum of vibration experienced during the operation of the blower.

3.16.1.3 Equilibrium. - Temperature and humidity equilibrium shall be reached within 10 minutes after closing the chamber door and starting the blower. Thereafter the temperature shall remain constant within $\pm 0.5^{\circ}$ C and the relative humidity within ± 2 percent over a 5 minute operating period.

3.17 Workmanship. - The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein:

3.6 Tropicalization
3.7 Finish
3.8 Hardware mounting
3.9 Marking
3.10 Caution notice
3.11 Cleaning
3.12 Wiring and cabling
3.13 Soldering
3.14 Interchangeability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see4. 2)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4. 3 and 4. 14).

(1) Procurement inspection of items before preparation for delivery. (see 4. 3).

(2) Procurement inspection of preparation for delivery (see 4. 14 and section 5)

4. 2 Preproduction Inspection: This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table II and the Group A inspection specified in tables III and IV. Other non-destructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, and (3) shock. Other preproduction inspection may precede, follow, or be inter-spersed between the foregoing.

Table II - Preproduction inspection

Inspection (for additional preproduction inspection see 4. 2)	Require, Par.	Test Par.
Vibration	3. 5	4. 10
Bounce	3. 5	4. 9
Moisture resistance	3. 6	4. 12
Temperature	3. 4	4. 13
Shock; drop	3. 5	4. 11

4. 3 Procurement inspection of items before preparation for delivery. - The minimum inspection performed by the contractor shall be as specified by (a) through (d) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its descretion--will verify by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A inspection as specified in 4. 3.

(b) When Standard MIL-STD-105 specified actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

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(c) Each unit which will be subjected to group A inspection shall be preconditioned after final assembly. (see par. 3.5.3)

4.3.1 Group A inspection. - This inspection (including sampling) shall conform to tables III and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the control unit operational inspection shall be last.

Table III - Government Group A inspection

Inspection	Require Para	Test Para	AQL% Major	Minor	Control
Visual and mechanical		4.5			
Control unit and chamber			1	4	10
Cable assemblies		4.5	2.5	4	10
Electrical					
Control unit and chamber		4.8.2	1.0		
Cable assemblies		4.7	1.0		
Control unit operation and continuity test	3.16.1.1	(4.8.2 (4.8.3	1.0		
Air circulation	3.16.1.2	4.8.1	1.0		
Dimensional	(3.2 (3.15	4.6	4.0		

4.3.2 Group B inspection. - Not required

4.3.3 Disposition of nonconforming product. - When defective sample units or failed inspection lots are submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Bounce preconditioning. - The unit shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.5 Visual and mechanical inspection. - Parts and equipment shall be examined for the defects listed in Standard MIL-STD-252 and in table IV below:

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Table IV Classification of Visual and Mechanical Defects

A. Chamber

1. Major

a. Corners not tight
b. Broken or cracked panels or corners
c. Delamination of plywood
d. Holes or gouges in plywood extending through more than one ply.
e. Splinters or protrusions of such a nature as to cause injury to using personnel.
f. Glass panel in door missing, broken or cracked.

2. Minor

a. Holes or gouges in plywood not extending through one ply.
b. Splinters or protrusions in case.

B. Finish

1. Major

a. Color or gloss not as specified.
b. Adhesion fails specified scrape or thumbnail test.
c. Scratches, cuts, abrasions, etc., causing exposure of bare metal or wood.

2. Minor

a. Adhesion - relative small amounts of flaking, peeling, or chipping
b. Scratches, cuts, abrasions, etc., causing exposure of primer.

3. Control

a. Color - difference in specified shade, hue or intensity.

C. Parts

1. Major

a. Missing or inoperative hinges, door fasteners, or doors.
b. Missing straps or retainers or clips.
c. Shelf missing.
d. Blower guard, reflector, or plastic cup missing or broken.
e. One or more leg support missing.
f. Control unit, psychrometer case, or cabinet assembly loose.

2. Minor

a. Blower guard, reflector or plastic cup not firmly attached.
b. Hinges, catches or fasteners difficult to operate.

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4.6 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.15. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.7 Testing of cable assemblies. -

4.7.1 Continuity. - Each conductor of the cable assemblies shall be tested for continuity and for correct connection to its terminations, using a low-voltage circuit with a suitable indicating device.

4.7.2 Dielectric strength and insulation resistance. - The test voltage shall be applied between each conductor, in turn, and the remaining conductors connected together. Either Method I or II may be used, at the option of the contractor.

4.7.2.1 Method I. - Dielectric strength shall be tested and insulation resistance measured in two steps, with step (b) immediately following step (a):

(a) Apply the test voltage for at least 5 seconds.

Test voltage shall be d.c. or sinusoidal a.c., and shall have the following: value:

A.C. test voltage = 2.00 times the rated a.c., of 1.41 times the rated d.c. voltage of the cable assembly, but at least 350 volts RMS.

D.C. test voltage = 2.83 times the rated a.c. or 2.00 times the rated d.c. voltage of the cable assembly, but at least 500 volts d.c.

(b) Apply at least 100 volts d.c., continuously, measuring the insulation resistance when the test instrument ceases to fluctuate and a constant or increasing reading of insulation resistance is attained. However, the period of time between application of the voltage and the reading of insulation resistance shall not exceed 1 minute. The test instrument shall be a galvanometer or other suitable instrument.

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4.7.2.2 Method II. - The two steps of method I may be combined, provided that a.d.c. test voltage complying with 4.7.2.1(a) is applied for at least 5 seconds but not more than 1 minute.

4.8 Performance tests. -

4.8.1 Air circulation. - With the blower motor on and the door of the temperature humidity chamber closed the rate of airflow across the ceiling of the chamber shall be not less than 10 mph and across the floor of the chamber not less than 3 mph.

4.8.2 Control unit continuity. - Wiring and other connections including ground connections to chassis, shall be checked for continuity and short circuits by a point-to-point test against Signal Corps Drawing Number SC-DL-21271.

4.8.3 Control Unit operation. - The relays shall be checked mechanically and electrically when the selector switch S1 is in the temperature, reference, humidity and automatic position in the following manner: (Note: Reference symbols, are as given in Drawing SC-DL-21271).

4.8.3.1 Temperature position. - With power switch S6 closed and selector switch S1 in temperature position no power shall be applied to relays K1 and K2 and the relays shall not be operative.

4.8.3.2 Humidity position. - With power switch S6 closed and selector switch S1 in humidity position, the circuit for relay K1 shall be completed and relay K-1 shall operate. This shall be electrically checked by obtaining continuity between contact 2 and 4 of the radiosonde test plug P1.

4.8.3.3 Reference position. - With power switch S6 closed and switch S1 in reference position, the circuit for relay K2 shall be completed and relay K2 shall operate. This shall be electrically checked by obtaining continuity between contact 1 and 4 of the radiosonde plug P1.

4.8.3.4 Automatic position. - With power switch S6 closed and selector switch S1 in the automatic position the one R.P.M. motor B2 shall start and actuate micro-switches S3 and S4 closing or opening the respective relays.

4.8.3.4.1 With the lever of switches S3 and S4 closed relay K2 shall become energized and operate. This shall be electrically checked by obtaining continuity between contact 1 and 4 of the radiosonde test plug P1. After switches S3 and S4 have operated in this position for 15 seconds, the lever of switch S3 shall enter the indent of the cam and switch S3 shall open.

4.8.3.4.2 With switch S3 opened and switch S4 closed relay K2 shall become energized and operate. This shall be electrically checked by obtaining continuity between contact, 1 and 4 of radiosonde test plug P1. After switch S3 has operated in this position for 15 seconds switch lever S3 shall emerge from indent of the cam.

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4.8.3.4.3 With switches S3 and S4 closed the relays shall be operated as in 4.8.3.4.1. After switches have operated in this position for 15 seconds switch lever S4 shall enter the indent of the cam and switch S4 shall open.

4.8.3.4.4 With the switch S4 open no relays shall be energized. No continuity shall exist between the contacts of the radiosonde test plug P1 when checked electrically. After switch S4 has operated in this position for 15 seconds switch lever S4 shall emerge from the indent of the cam.

4.9 Bounce test. - The equipment, in its carrying case, shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm $\pm$ 1 percent, for a total of 3 hours. During each 1/2-hour period of the test the carrying case (with contained equipment) shall rest on a different face.

4.10 Vibration test. - The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.5.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecing mechanical vibration through electrical output indications.

4.11 Shock Test, Drop. - The equipment, in its carrying case, shall be subjected to the drop test in accordance with the following conditions:

a. The floor or barrier receiving the impact shall be of 2 inch fir, backed by concrete on a rigid steel frame.

b. The drops shall be made from a quick release hook, or a pendulum tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal.

c. The drops shall be made from a height of 30 inches.

d. The equipment shall be dropped on each corner, for a total of 8 drops.

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4.12 Moisture-resistance test for equipment. - The equipment shall be tested as follows:

4.12.1 Test conditions. -

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Per form last two cycles with the chassis removed from its enclosure.

4.12.2 Test. -

(a) Dry at 130° ±5° for 24 hours.

(b) Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.12.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.12.3 shall be made during the periods shown in the standard and shall comply with 4.12.3.

((e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.12.3.

4.12.3 Performance. - Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	Measurement	Performance during cycling
4.8.1	Air circulation	Full specification performance
4.8.3	Control unit operation	Full specification performance

4.12.4 Failure. - If any equipment fails to meet the performance specified in 4.12.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.12.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.12.2(e), it does not pass the test.

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4.13 Temperature test. - The equipment shall be subjected to the temperature cycle shown on standard MIL-STD-169. Measurements shall be taken as follows:

Paragraph	Steps	Measurement	Test Result
4.8.1	1, 4, 8, 10	Air Circulation	Full Specification Requirements
4.8.2	1, 4, 8, 10	Control Unit Continuity	Full Specification Requirements
4.8.3	1, 4, 8, 10	Control Unit Operation	Full Specification Requirements

4.14 Procurement inspection of preparation for delivery. - The minimum inspection performed by the contractor shall be as specified in 4.14.1 and4.14.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government - at it discretion - will verify by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specified actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.14.1 Preservation and packaging. - Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table V.

TABLE V

PRESERVATION, PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Packaging material damaged.
3. Incorrect packaging method applied.	3. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible.
4. Cushioning or padding omitted.	
5. Cushioning inadequate for the physical and mechanical protection of the item.	
6. Unsealed, punctured or improperly sealed barrier bag, wrap or envelope.	
7. Stock number omitted, incorrect, or illegible.	
8. Nomenclature omitted, incorrect, or illegible.	
9. Marking of quantity of items in package omitted, incorrect, or illegible	
10. Different stock numbered items in the same unit package.	

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4.14.2 Packing and marking of exterior containers. - Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table VI.

TABLE VI

PACKING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1 Unsealed carton
2. Quantity in pack not as specified	2. Defective taping or sealing of carton.
3. Gross weight in excess of specified amount.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Box closure not as specified.	4. Any item of required marking information other than 10 thru 16 listed under major defects incorrect, omitted, or illegible.
5. Type, grade, class, and style of the shipping container not as specified.	
6. Strapping omitted.	
7. Strapping inadequate or incorrectly applied.	
8. Items not adequately blocked, braced, or cushoned within the shipping container to prevent movement or damage.	
9. Shipping documents or packing list omitted.	
10. Stock number omitted, incorrect, or illegible.	
11. Nomenclature omitted, incorrect, or illegible	
12. Marking of quantity of items in pack omitted, incorrect, or illegible.	
13. Destination marking omitted, incorrect, or illegible.	
14. Service designation (color marking) omitted.	
15. Specified special marking and labeling such as NAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.	
16. Overseas code marking omitted, incorrect, or illegible.	

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4.14.3 Inspection lot. - A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.14.4 Procedure in case of failure. - If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.14.5 Disposition of nonconforming product. - Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall be come the property of the Government.

4.14.6 Rough handling test (preparation for delivery). - When rough handling test is required by the contract (see 6.2(d)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

a. Visual and Mechanical Inspection (see Par 4.5)

b. Performance Tests (see Par. 4.8.3)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. - Radiosonde Baseline Check Set AN/GMM-1( ) shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described herein.

5.1.1.1 Technical manual. - Each technical manual shall be packaged Method IC-3 as follows: Inclose each technical manual within a close-fitting, type II class b bag, conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 Lamps and fuses (spares). - Spare lamps and fuses shall be packaged Method III as follows: Items bearing the same stock numbers shall be placed together within a wrap of flexible corrugated fiberboard, type I, conforming to Specification LLL-F-291. Secure the wrap with gummed paper tape, class 1, conforming to Specification UU-T-111.

5.1.1.3 Thread spool and wicks. - Each item shall be packaged individually Method IC-3 as follows: Inclose each item within a close-fitting type I, class b bag, conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.4 Radiosonde Baseline Check Set AN/GMM-1( ) and spare parts in carrying case. - Each set shall be packaged Method IC-5 as follows: Stow components of set together with space parts packaged as specified in 5.1.1.2 and 5.1.1.3 within their respective compartments of carrying case. Close case and secure

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all fastenings. Cushion each case on all surfaces with cells or pads or both, fabricated of corrugated fiberboard having a minimum dry bursting strength of 275 pounds and designed to absorb the shock of impact, encountered in handling and transit. Cells and pads shall have a water resistance equal to that of the box in which they are used. Place each cushioned check set in carrying case within a close-fitting, style RSC, class 2, W5c, fiberboard box, conforming to Specification PPP-B-636. Box closure shall be effected by sealing all joints and seams with water resistant, pressure-sensitive tape, conforming to Specification PPP-T-76.

5.1.2 Level C. - Radiosonde Baseline Check Set AN/GMM-1( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage, during direct shipment to the first receiving activity.

5.2 Packing. -

5.2.1 Level A. - Radiosonde Baseline Check Set AN/GMM-1( ) packaged as specified in 5.1.1.4, shall be packed within a class 2, style 4, nailed wood box, conforming to Specification PPP-B-621. Fabricate the box to fit contents snugly. Place technical manuals, packaged as specified in 5.1.1.1, between the contents and the lid of the shipping container. Closure shall be in accordance with the box specification.

5.2.1.1 Metal strapping. - Shipping containers shall be strapped in conformance with the requirements of the appendix of the container specification, only for direct shipment to ports.

5.2.2 Level B. - Radiosonde Baseline Check Set AN/GMM-1( ) shall be packed as specified in 5.2.1 except that the style 4 nailed wood box shall be class 1, conforming to Specification PPP-B-621.

5.2.3 Level C. - Radiosonde Baseline Check Set AN/GMM-1( ) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 Consolidated Freight Classification Rules 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 Marking. - Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - Radiosonde Baseline Check Set AN/GMM-1( ) is a portable chamber used to establish stable temperature and humidity conditions for testing radiosondes AN/AMT-2 and AN/AMT-4.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

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(d) Preproduction inspection

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample.

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on Drawing SC- C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in lubrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(e) Marking and shipping of samples.

(f) Place of final inspection

(g) Technical literature required

(h) Quantity of tools and running spare parts required.

(i) Submisstion of the statement of treatment referenced in 3. 6 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6. 3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example; AN/GMM-1W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (see 1. 1)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture use, or sell any patented invention that may in any way be related thereto.

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